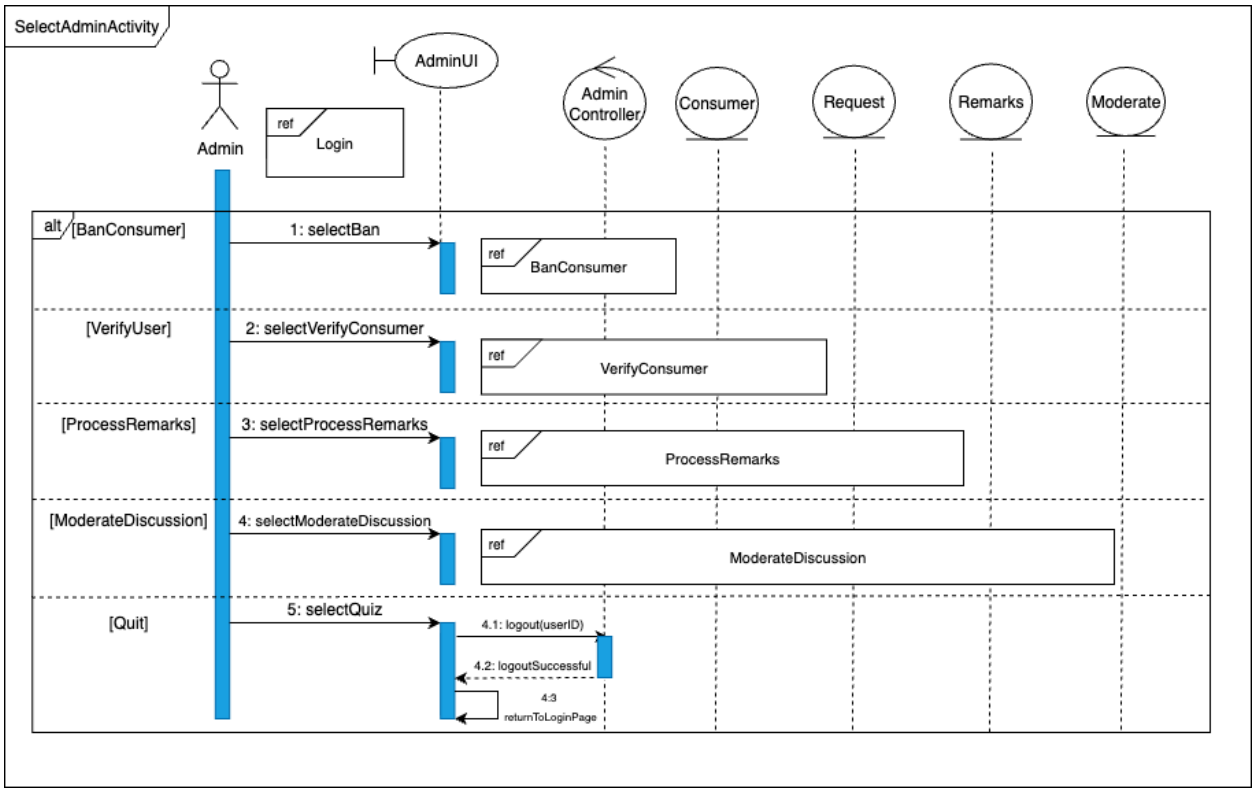
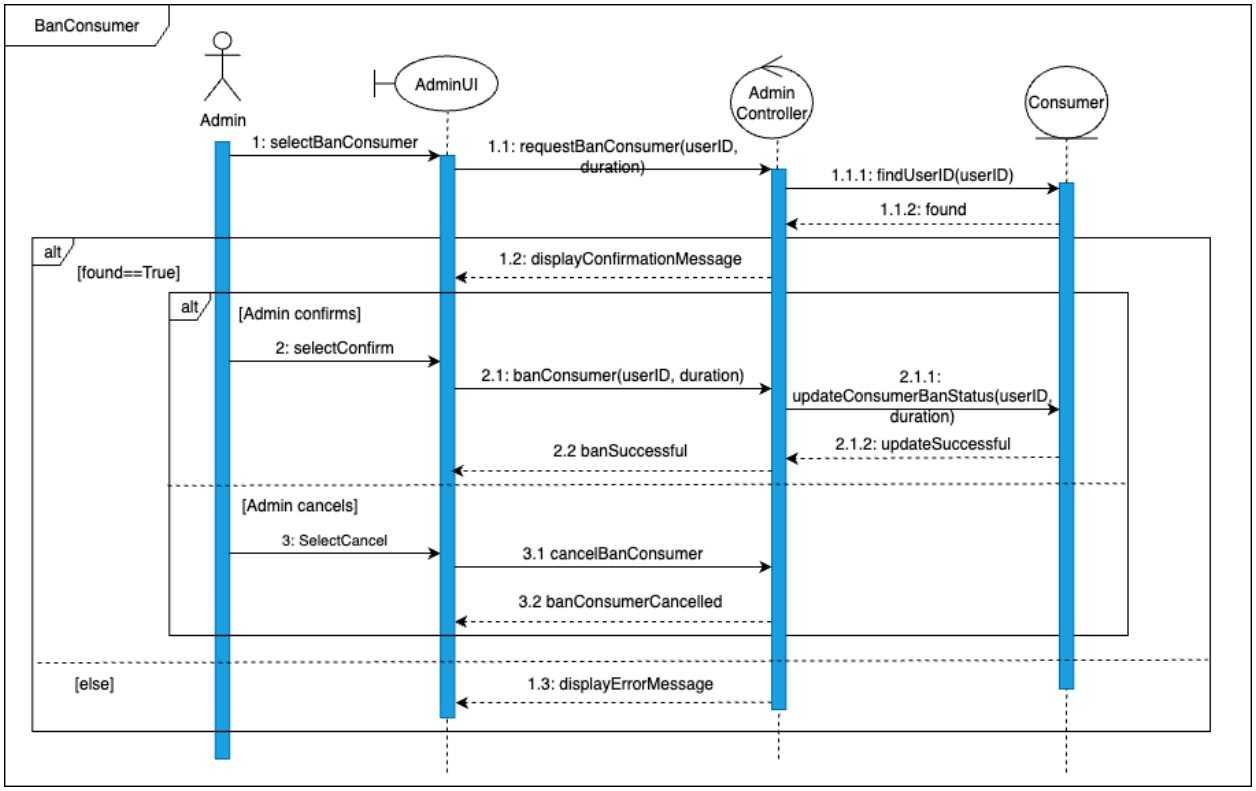
A. Sequence Diagram

I. For Use Cases Under I (Admin)

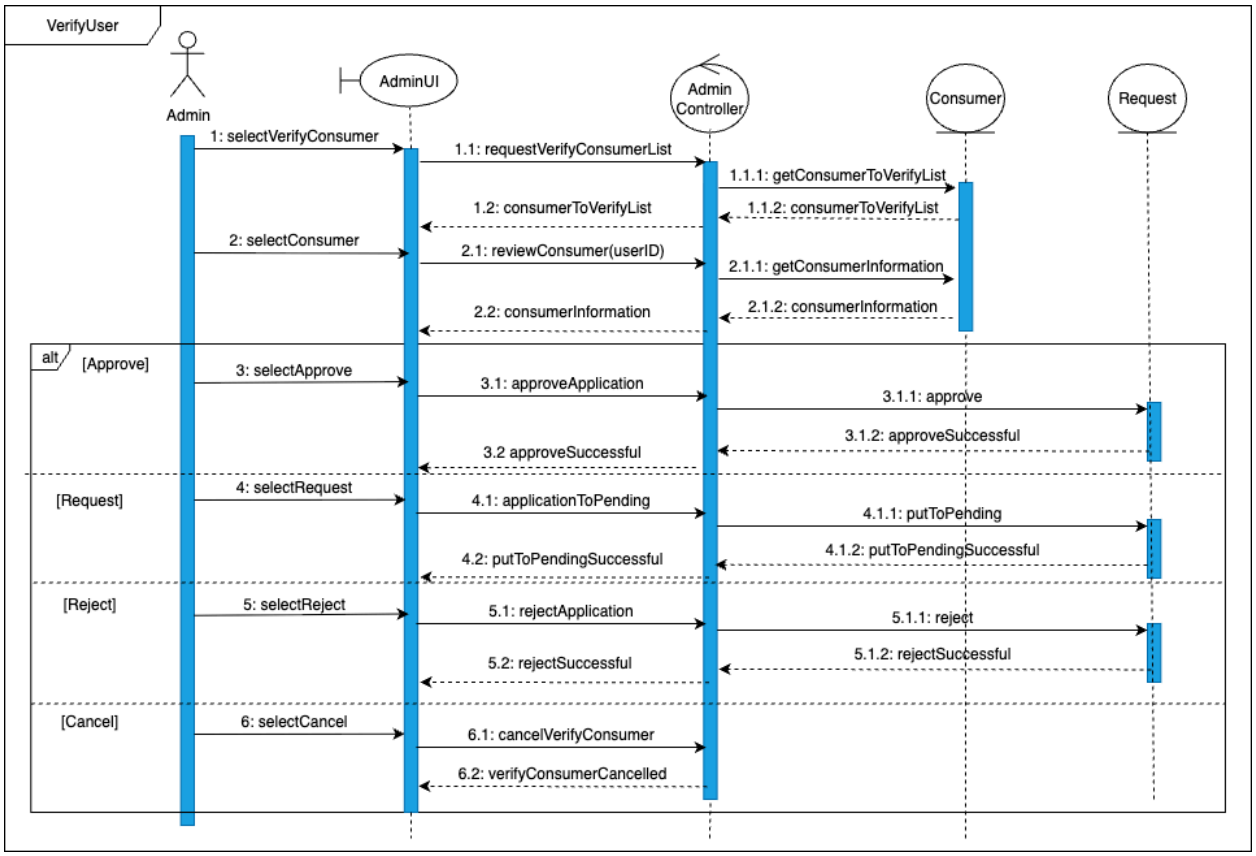
I.I SelectAdminActivity



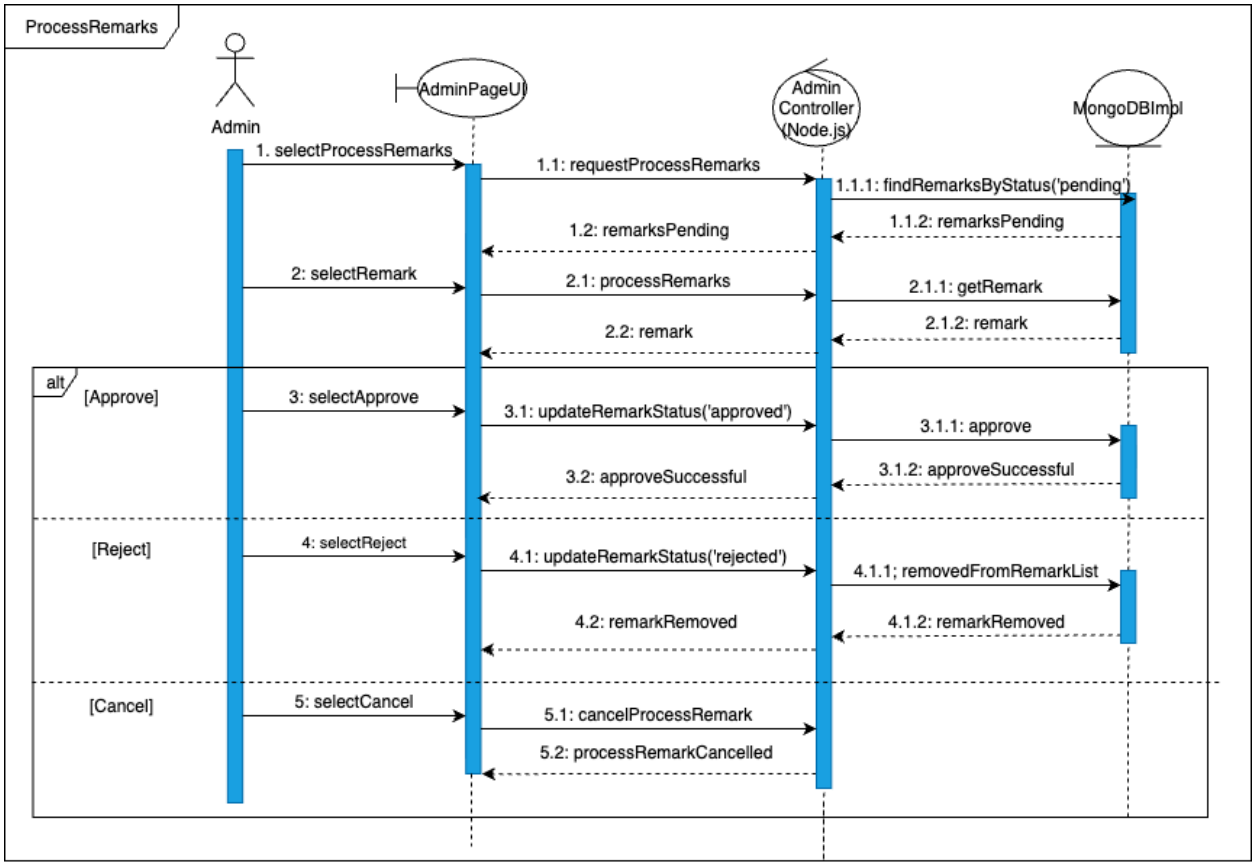
I.II BanConsumer



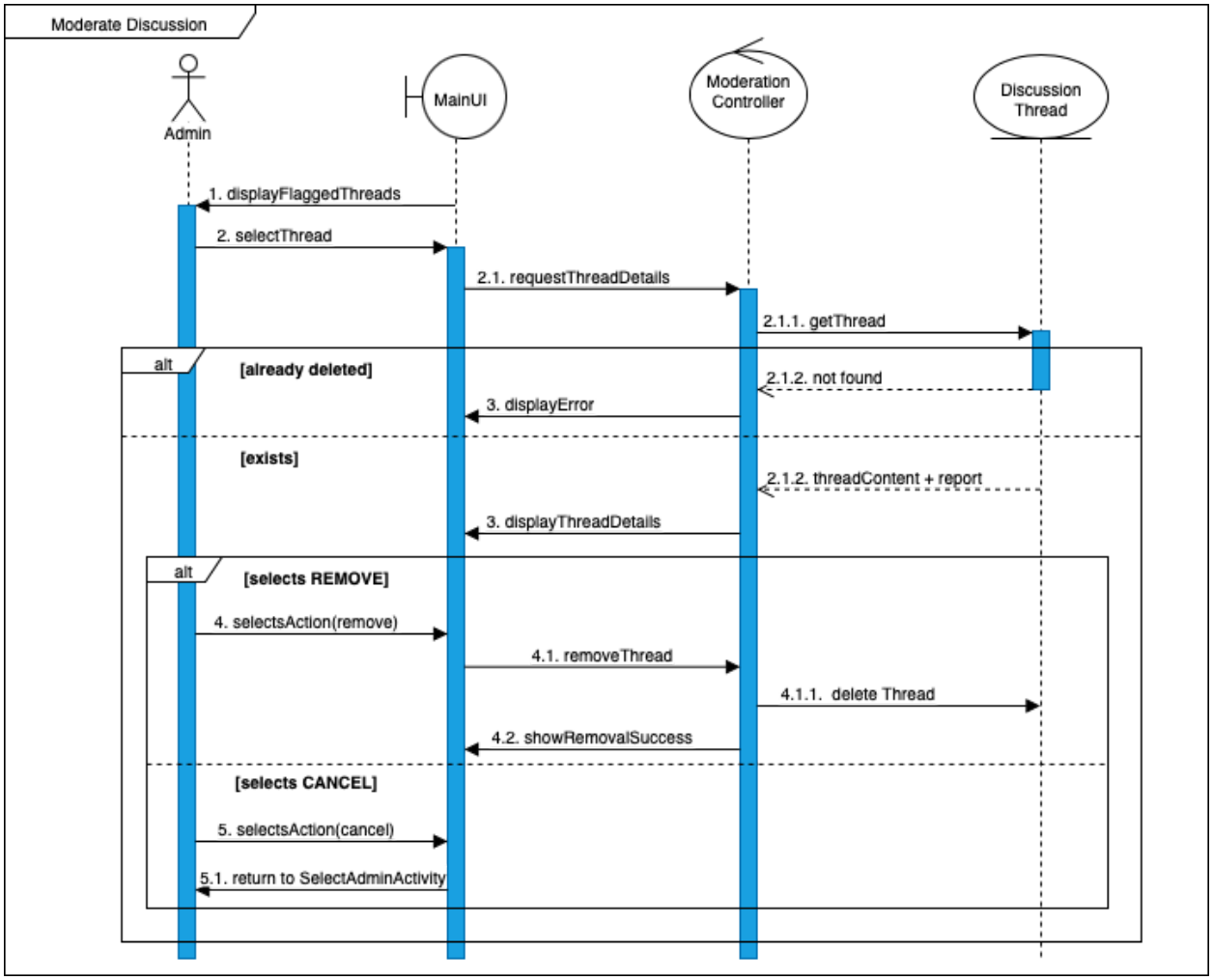
I.III VerifyConsumer



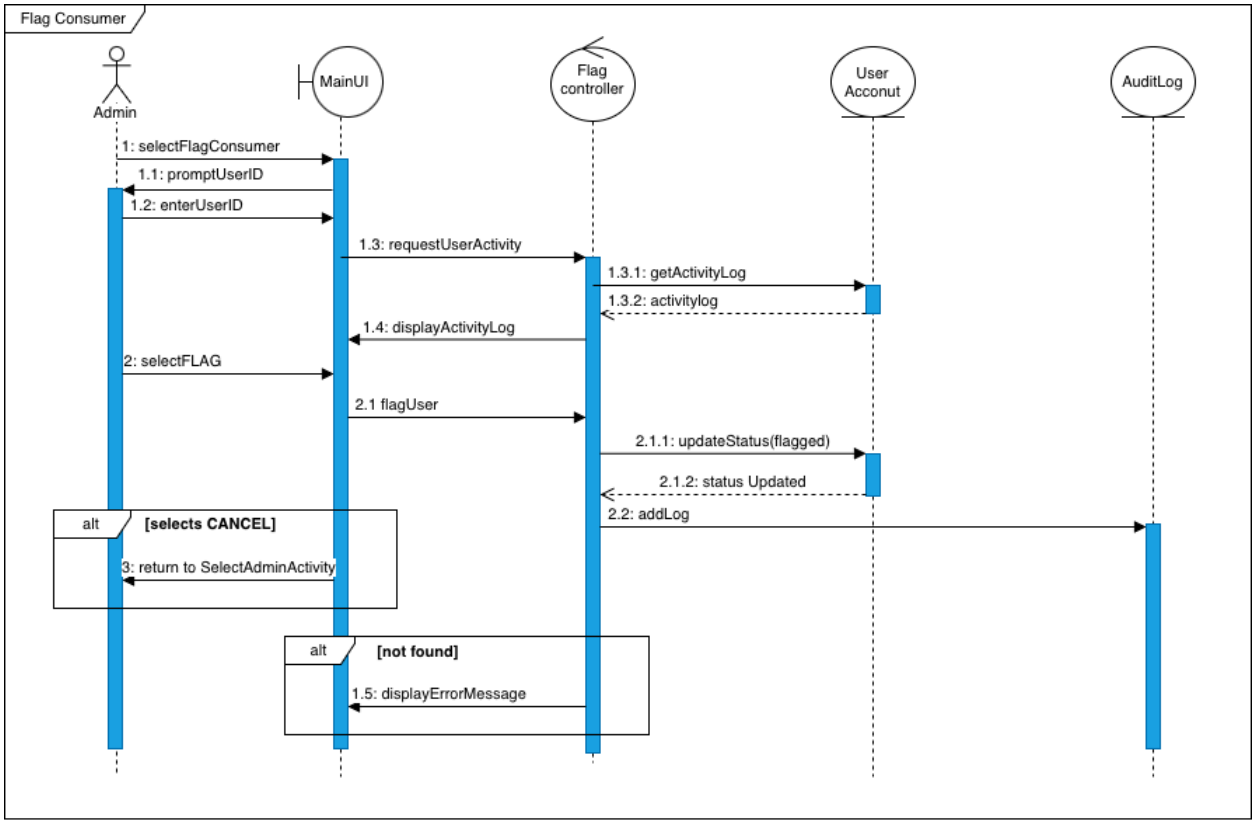
I.IV ProcessRemarks



I.V Moderate Discussions

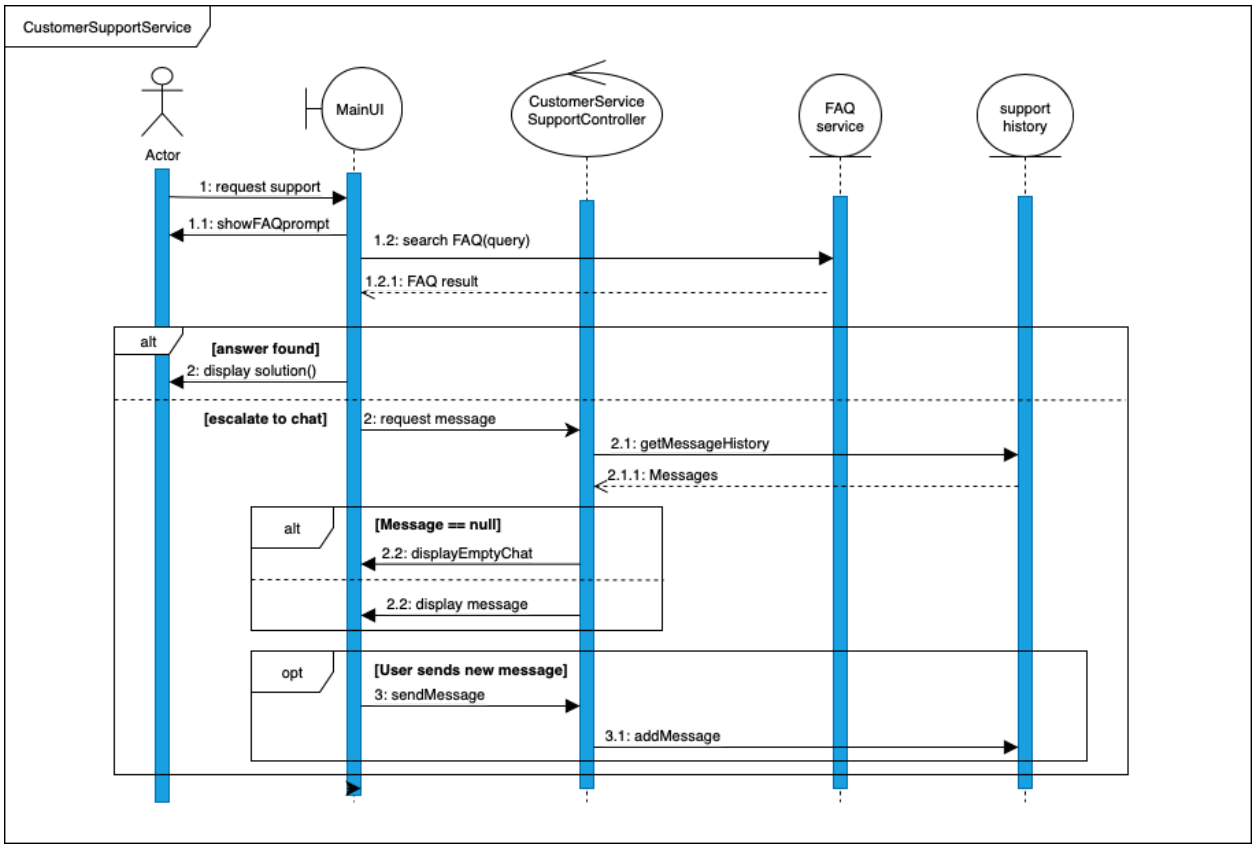


I.VI Flag Consumer



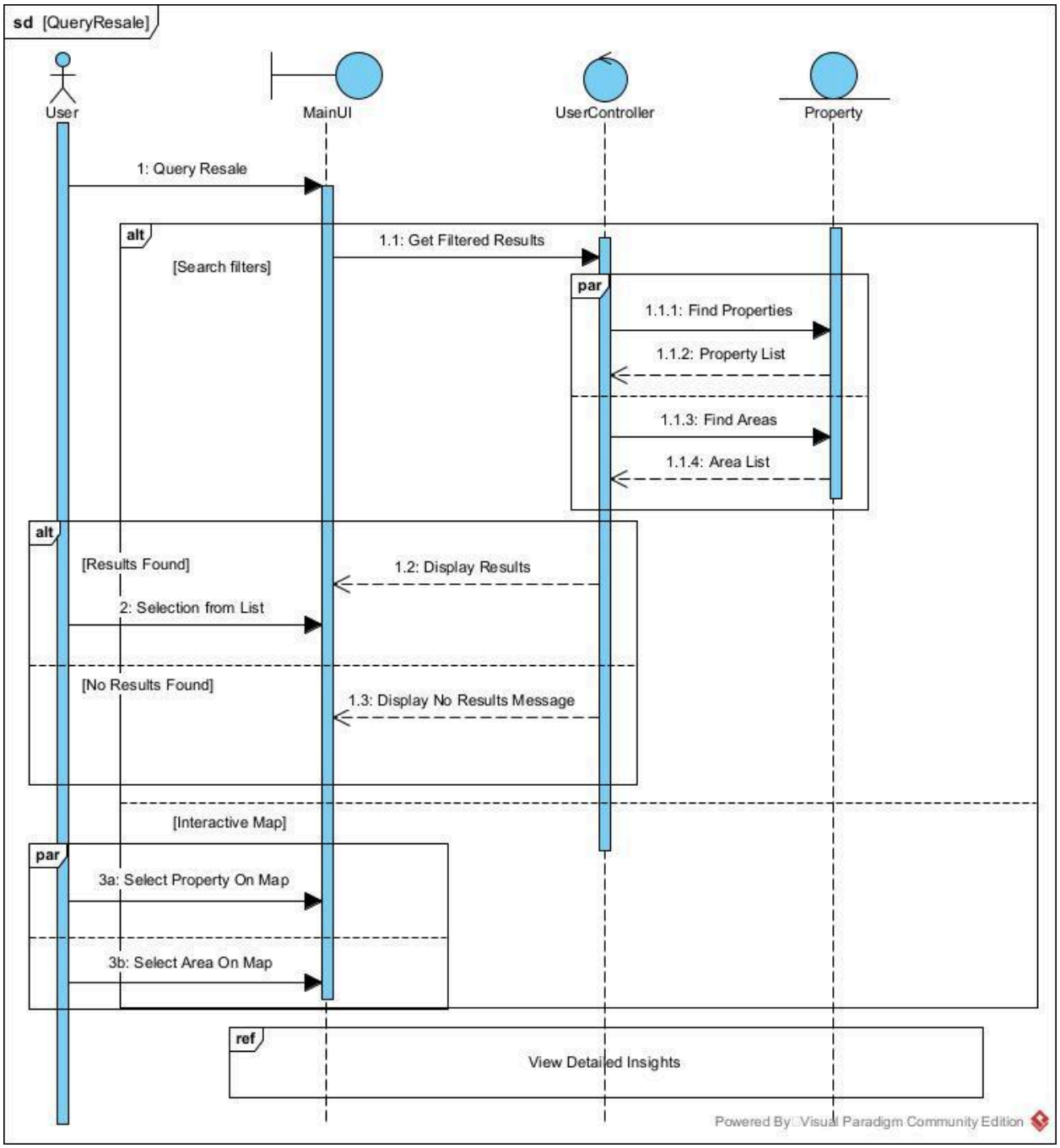
II. For Use Cases Under II

II.I Customer Service Support

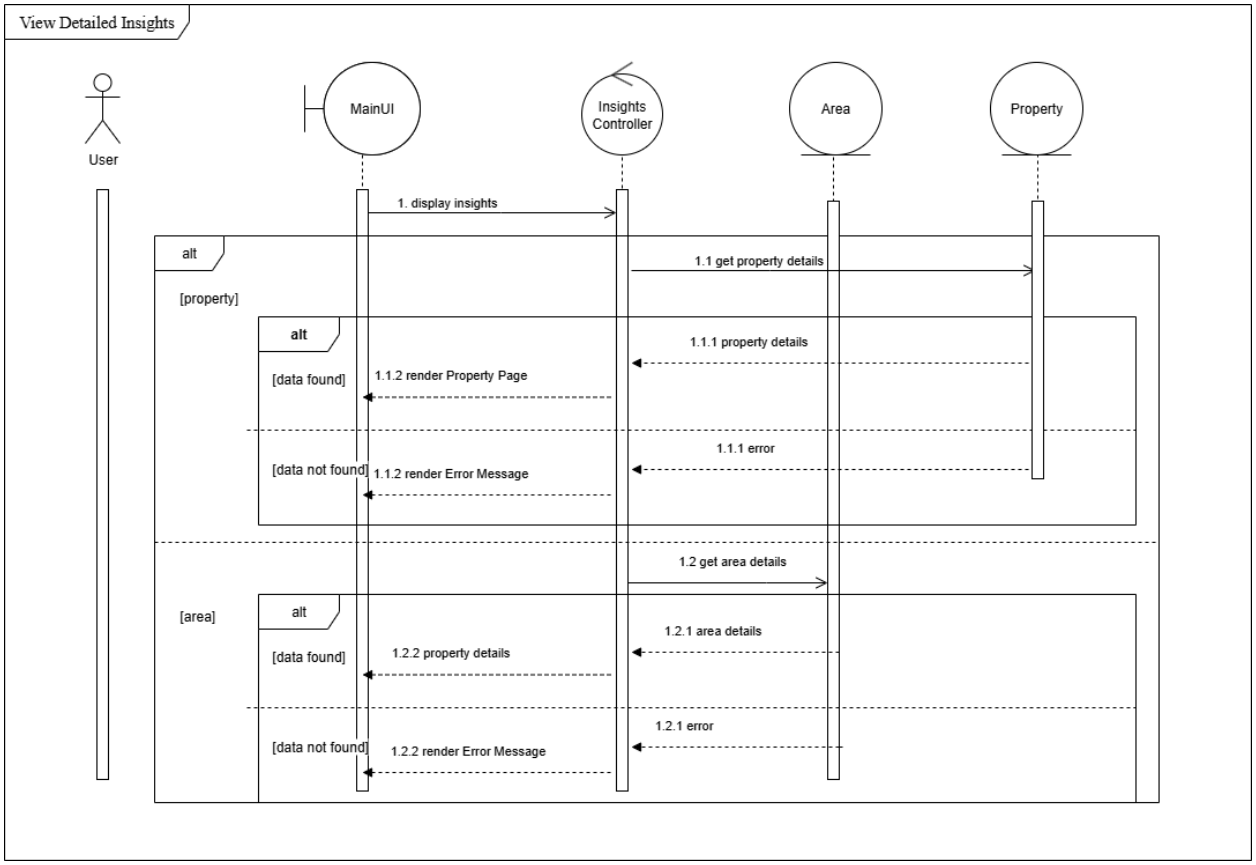


III. For Use Cases Under III

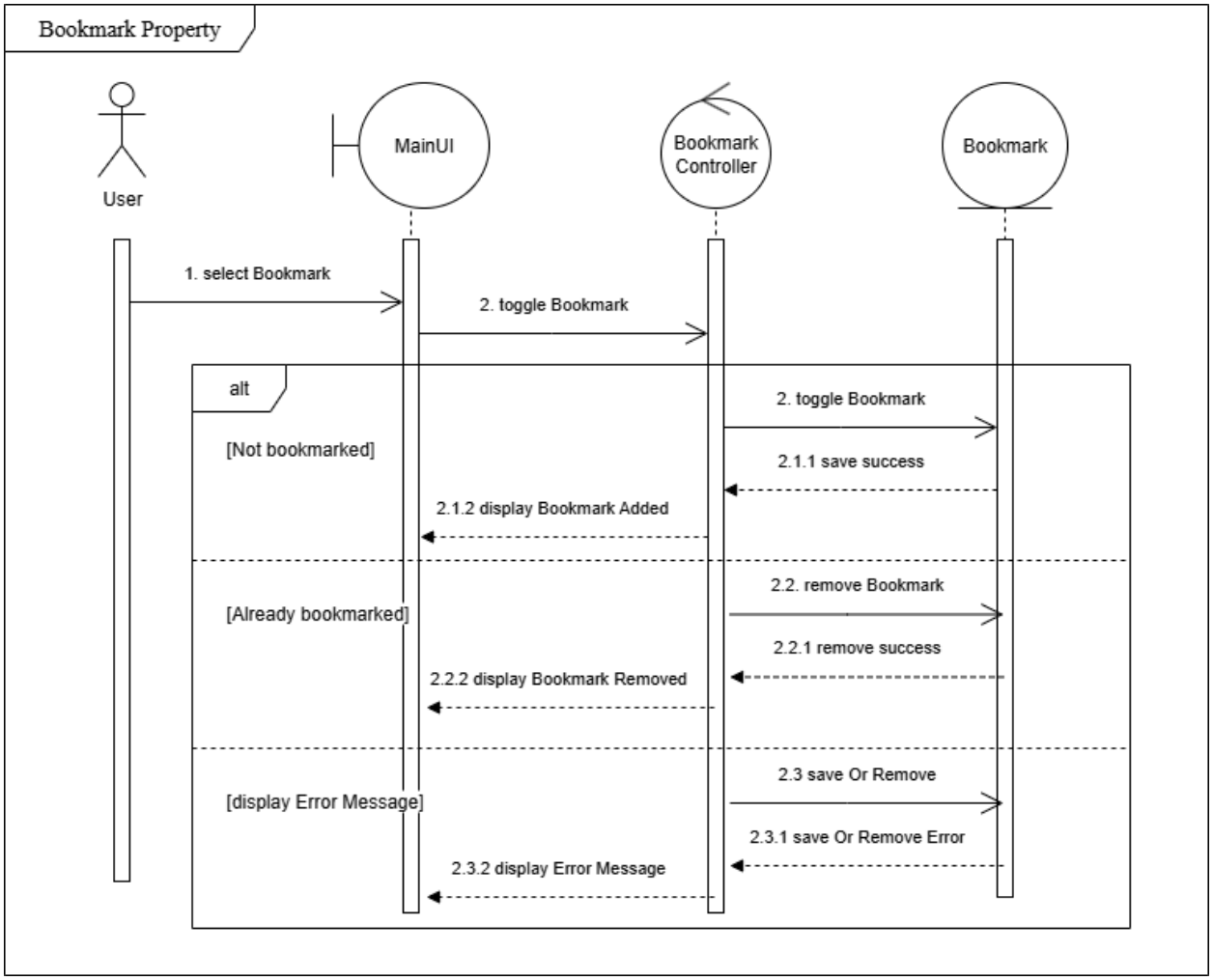
III.I QueryResale



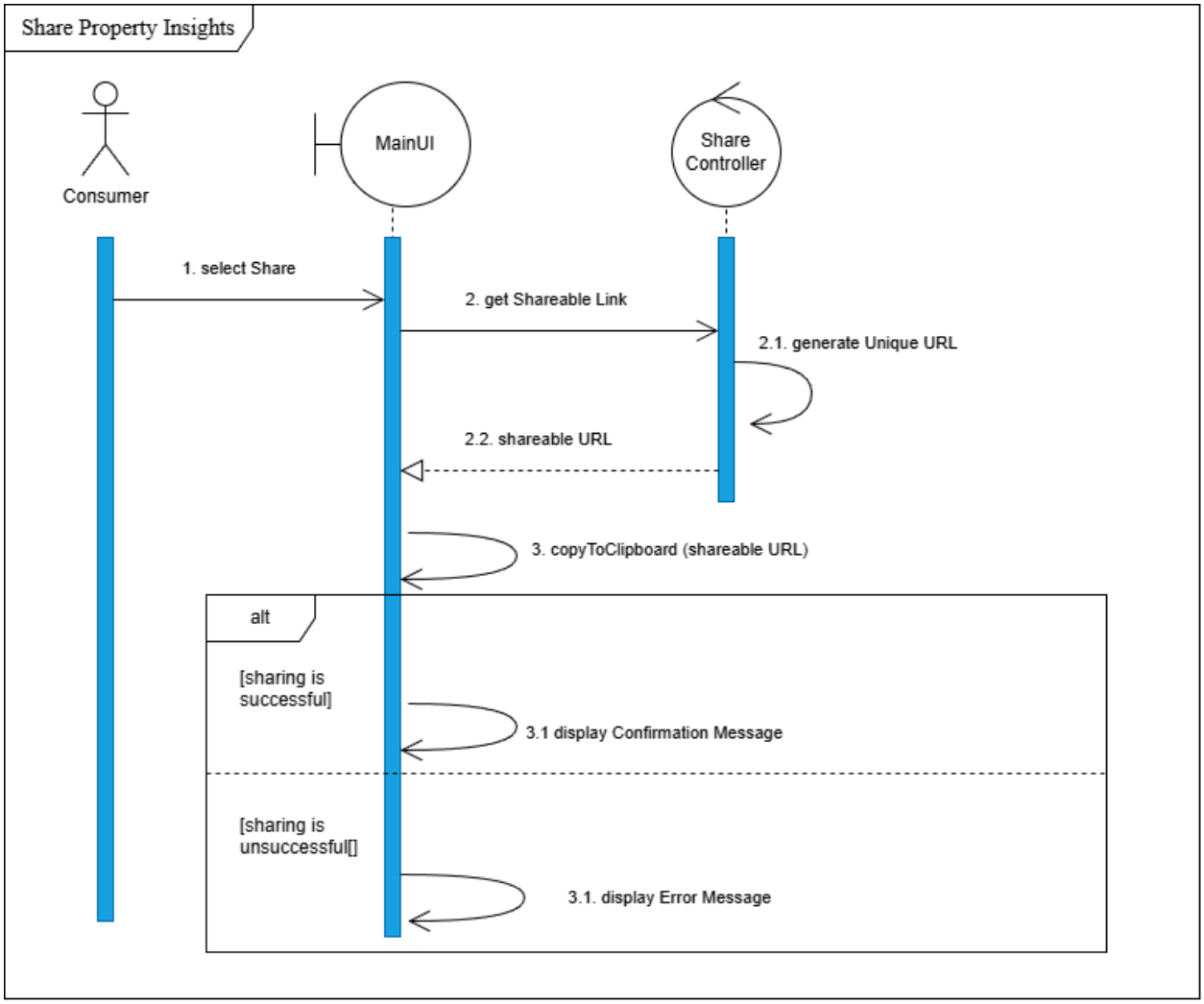
III.II ViewDetailedInsights



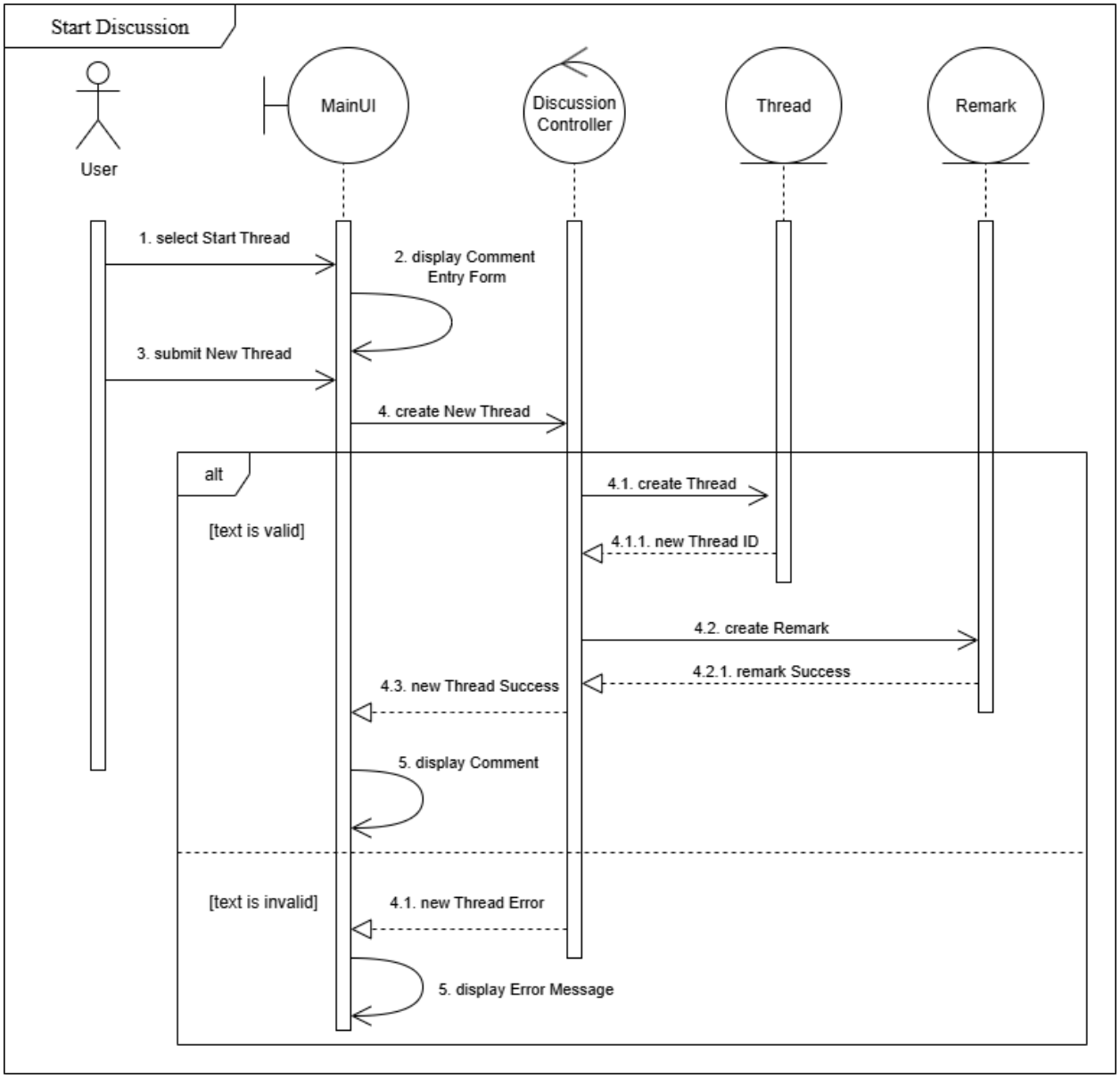
III.III BookmarkProperty



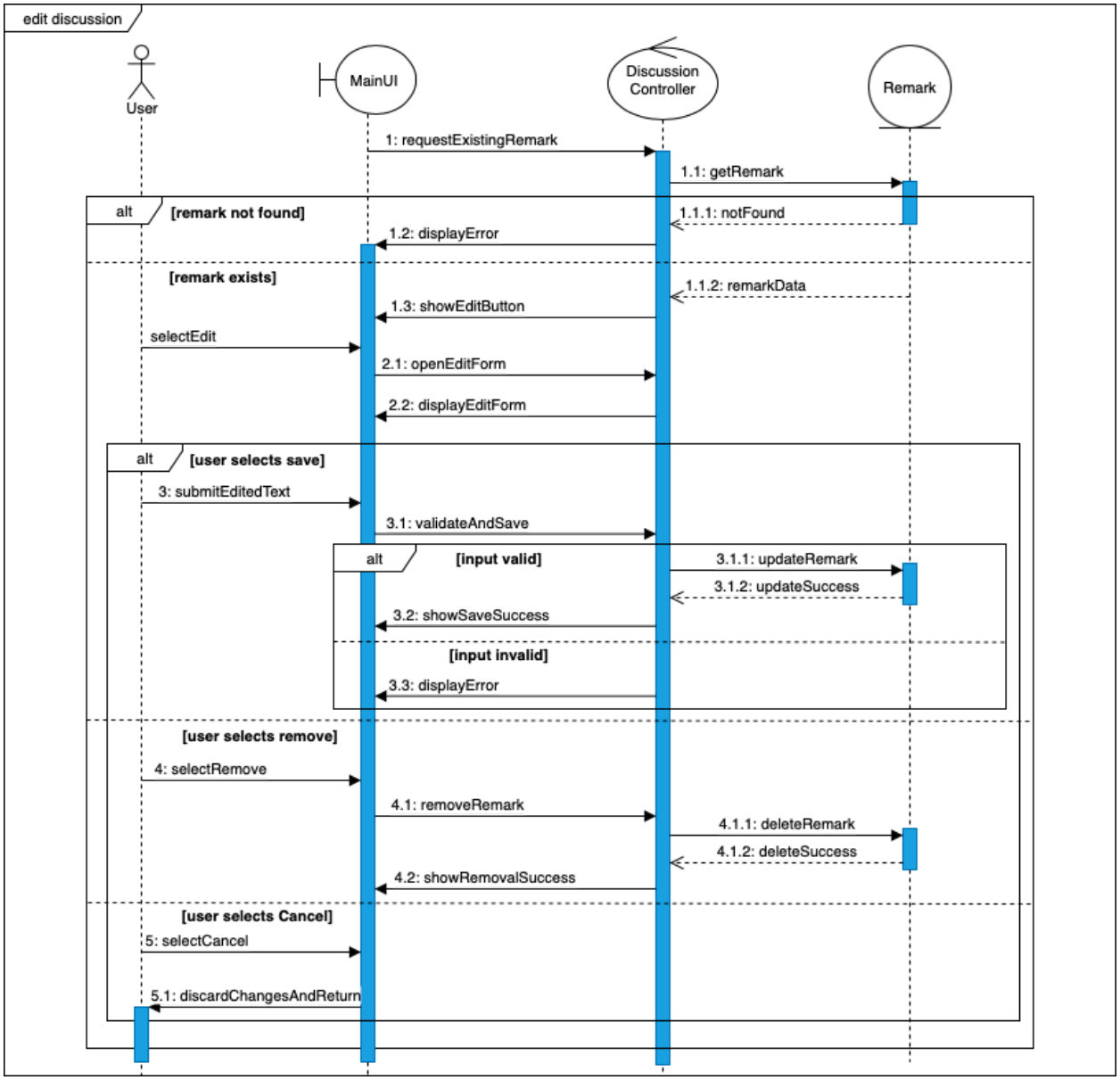
III.IV SharePropertyInsights



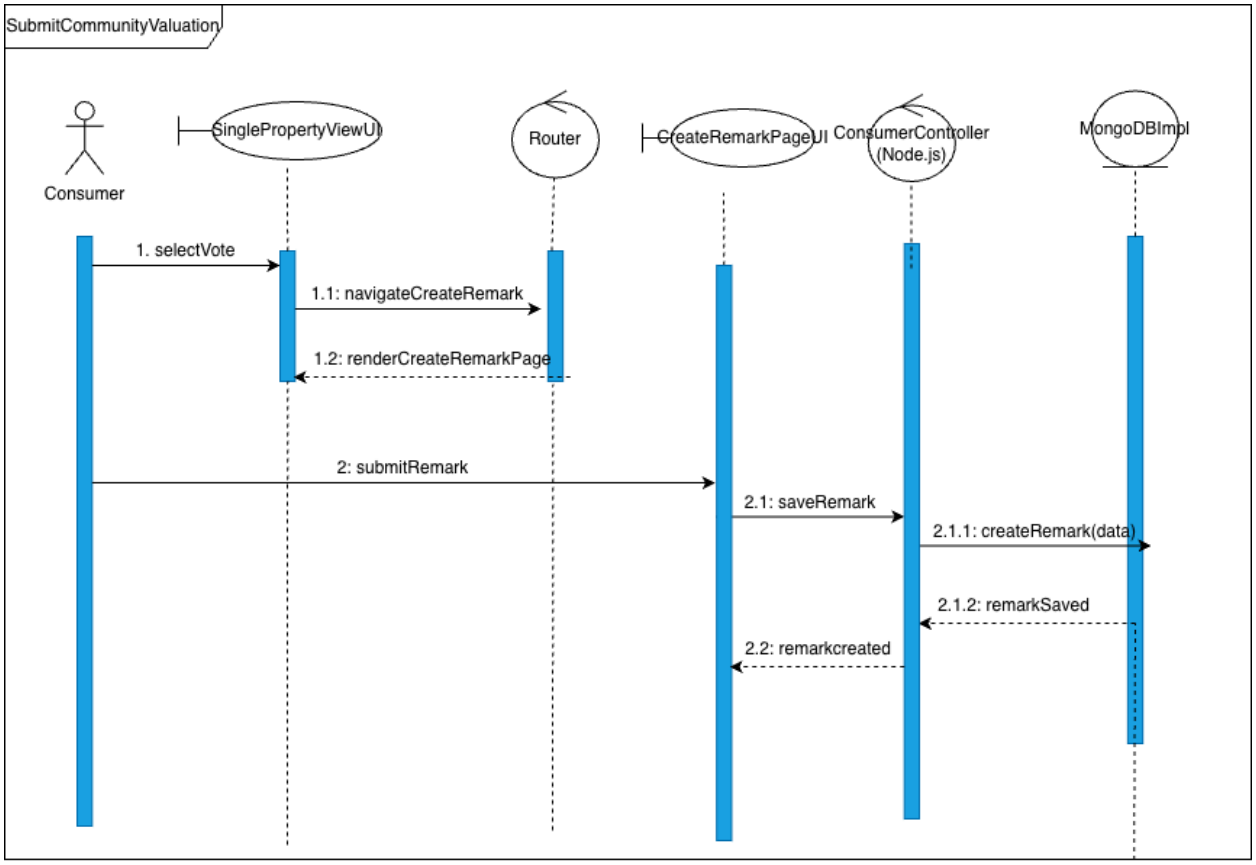
III.V StartDiscussion



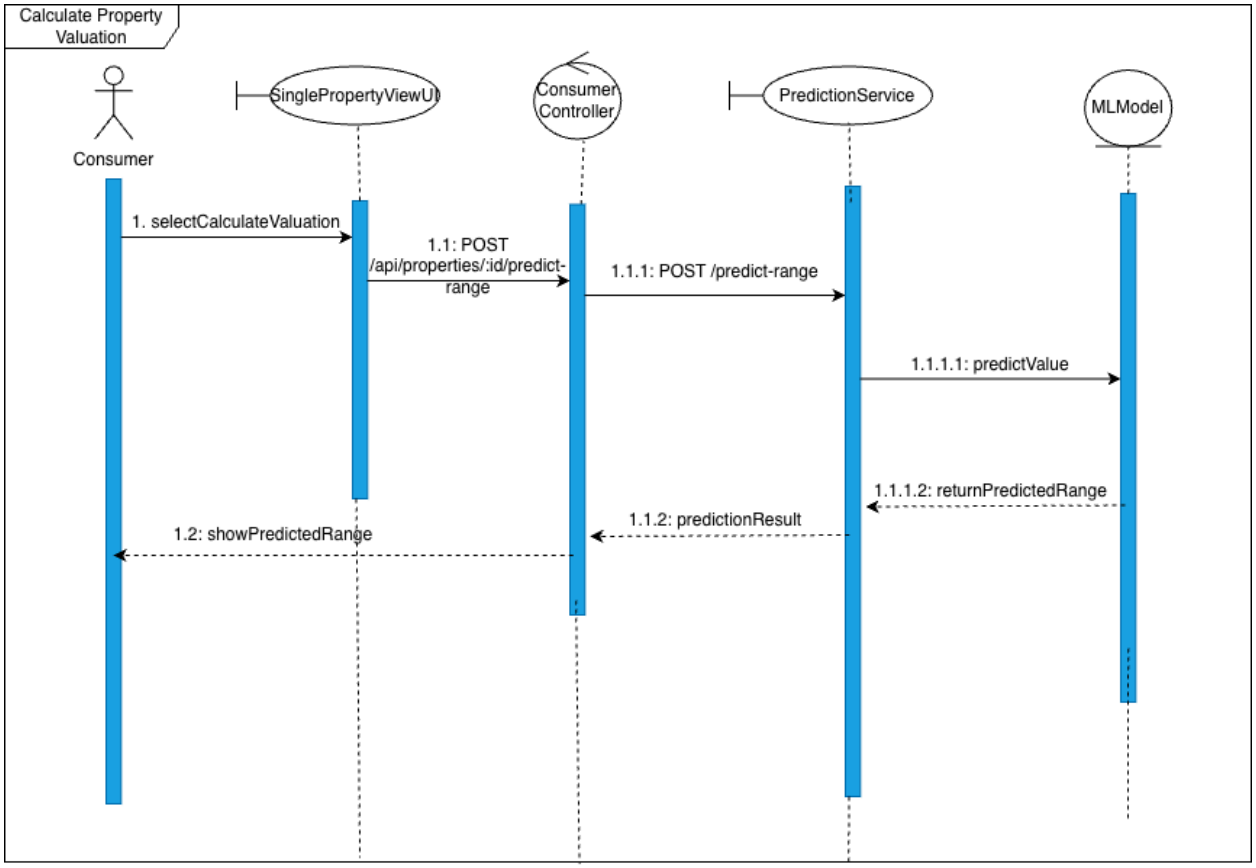
III.VI Edit Discussion



III.VII Submit Community Valuation

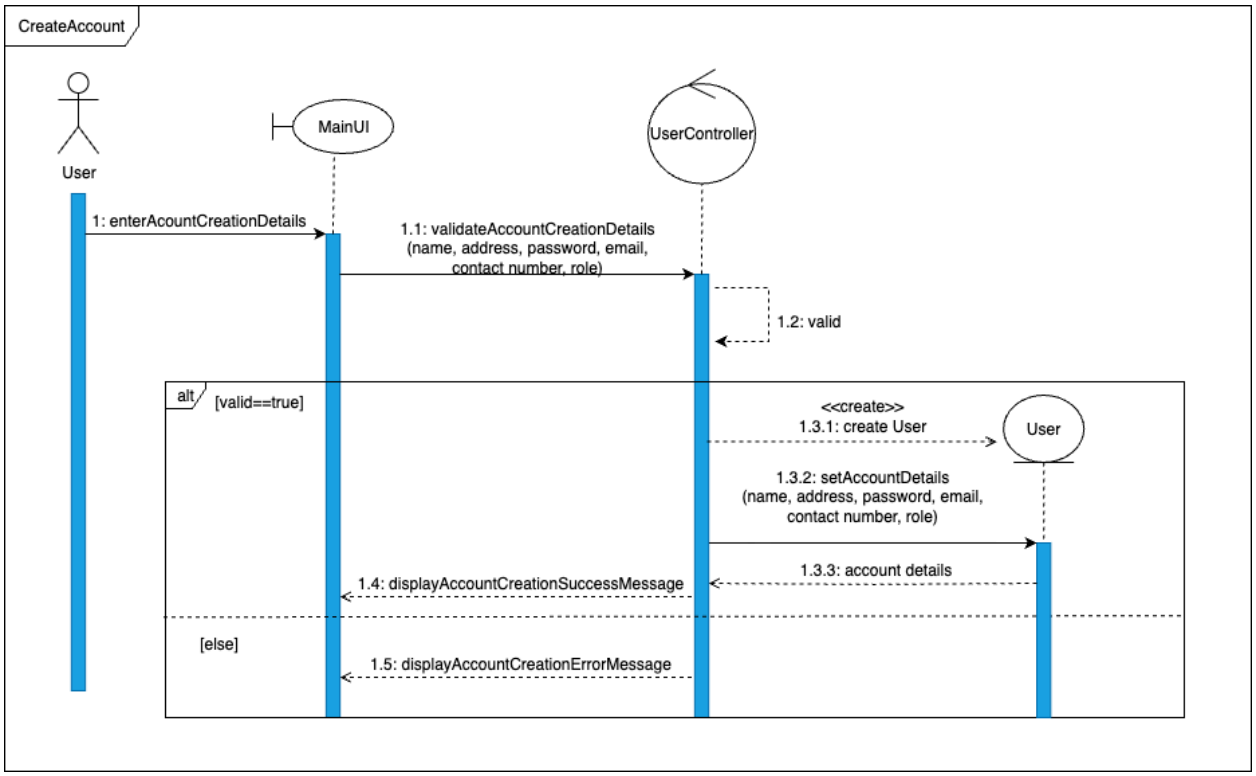


III.VIII Calculate Property Valuation

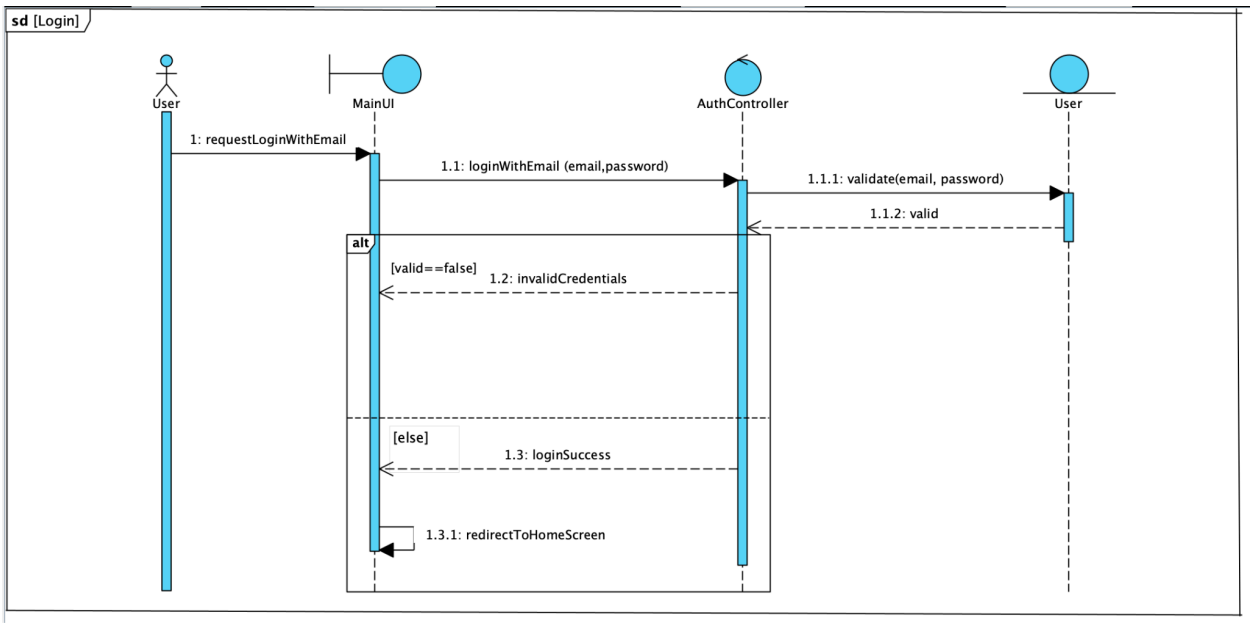


IV. For Use Cases Under IV

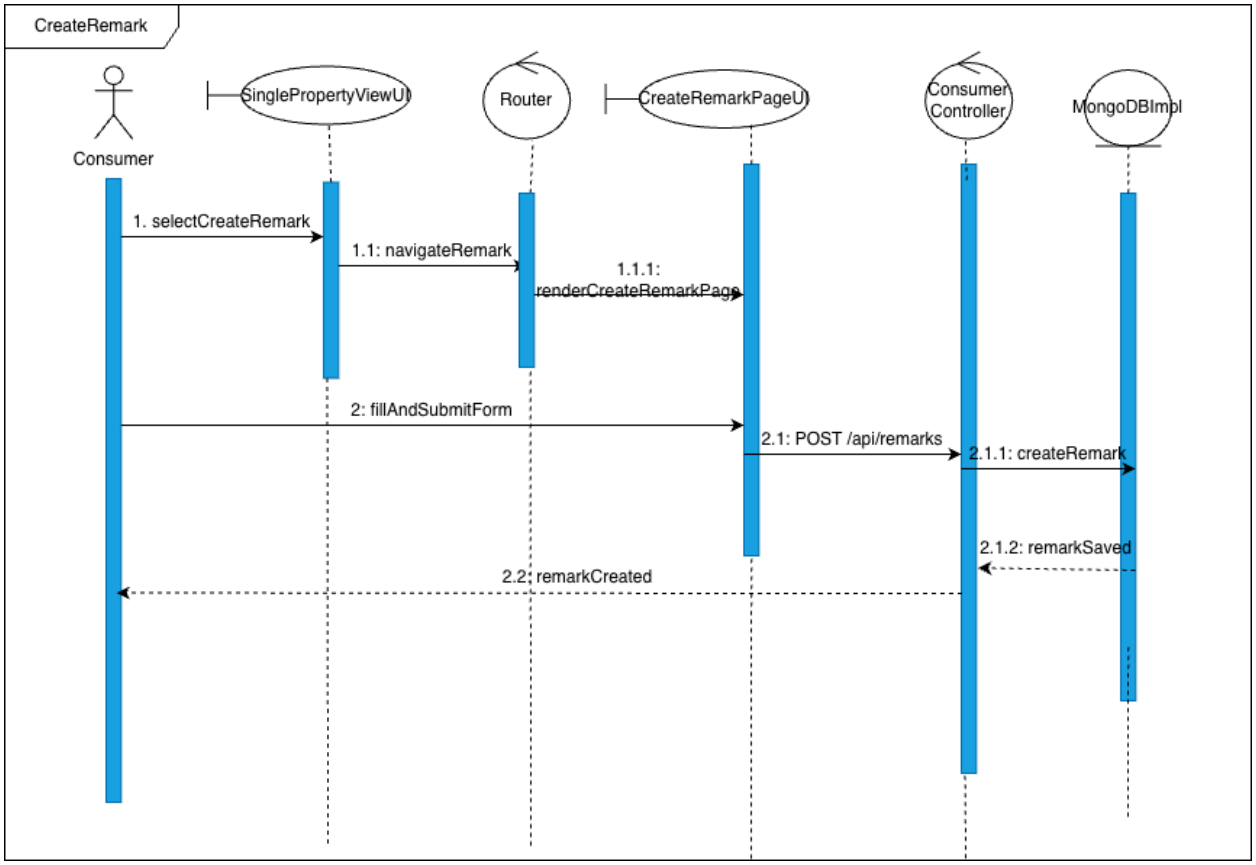
IV.I Create Account



IV.II Login


sd [Login]
User
MainUI
AuthController
User
1: requestLoginWithEmail
1.1: loginWithEmail (email,password)
1.1.1: validate(email, password)
1.1.2: valid
alt
[valid==false]
1.2: invalidCredentials
[else]
1.3: loginSuccess
1.3.1: redirectToHomeScreen

IV.III Create Remark



I. For Functional Requirement #1 (Admin)

I.I SelectAdminActivity

Use Case ID:	#1-1		
Use Case Name:	SelectAdminActivity		
Created By:	Tang Xinyue	Last Updated By:	Tang Xinyue
Date Created:	1st September 2025	Date Last Updated:	15st November 2025

Actor:	Admin
Description:	Allows Admin to select actions to take
Preconditions:	Admin is logged in and is authenticated
Postconditions:	None
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The system uses the included use case Login to verify the Admin. 2. On successful login, the system prompts the Admin to select the desired activity: BAN, VERIFYUSER, MODERATE, REMARKS, or QUIT. 3. If the Admin selects the activity BAN, then the Admin uses the included use case BanUser to ban a user. 4. If the Admin selects the activity VERIFYUSER, then the Admin uses the included use case VerifyUser to approve or reject a user's application for priority. 5. If the Admin selects the activity REMARKS, then the Admin uses the included use case ProcessRemarks to decide whether a review is to be removed. 6. If the Admin selects the activity MODERATE, then the Admin uses the included use case ModerateDiscussion to moderate a thread.. 7. If the Admin selects the activity FLAG, then the Admin uses the included use case FlagUser to flag a user. 8. If the Admin selects the activity QUIT, the system returns to the login screen.
Alternative Flows:	None
Exceptions:	None
Includes:	1. BanUser 2. VerifyUser 3. ProcessRemarks 4. ModerateThread 5. FlagUser
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

I.II BanUser

Use Case ID:	#1-2		
Use Case Name:	BanUser		
Created By:	Tang Xinyue	Last Updated By:	Tang Xinyue
Date Created:	3rd September 2025	Date Last Updated:	15st November 2025

Actor:	Admin
Description:	Allows Admin to ban a User by their user ID for a specified duration
Preconditions:	Admin is logged in and authenticated
Postconditions:	User is either banned or not banned by the admin
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. Admin inputs the user ID of the user that the Admin wants to ban. 2. Admin inputs the duration of the ban. 3. Admin confirms the ban by selecting CONFIRM.
Alternative Flows:	AF-S3: If Admin selects the CANCEL button 1. The system will return to the SelectAdminActivity.
Exceptions:	1. If the user ID is not found, the system shall display an error message and prompt the Admin again.
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

I.III VerifyUser

Use Case ID:	#1-3		
Use Case Name:	VerifyUser		
Created By:	Tang Xinyue	Last Updated By:	Tang Xinyue
Date Created:	3rd September 2025	Date Last Updated:	15st November 2025

Actor:	Admin
Description:	Allows Admin to verify a User by their user ID.
Preconditions:	Admin is logged in and is authenticated
Postconditions:	The user's application is either approved, pending or rejected
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The system displays the list of pending user verification requests. 2. Admin selects a user application from the list. 3. The system displays the user's submitted personal information. 4. Admin selects one of the following: ○ APPROVE → User is verified. ○ REJECT → Application is rejected.
Alternative Flows:	AF-1: If Admin selects CANCEL, the system returns to the SelectAdminActivity
Exceptions:	None
Includes:	None
Special Requirements:	Verification actions must be logged for auditing
Assumptions:	None
Notes and Issues:	None

I.IV ProcessRemarks

Use Case ID:	#1-4		
Use Case Name:	ProcessRemarks		
Created By:	Tang Xinyue	Last Updated By:	Tang Xinyue
Date Created:	1st September 2025	Date Last Updated:	15st November 2025

Actor:	Admin
Description:	Allows Admin to process a pending remark, approving it for public view or rejecting it.
Preconditions:	Admin is logged in and is authenticated
Postconditions:	The remark's status is set to "approved" or "rejected.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The ResaleSense lists the remarks that have been sent. 2. Admin selects the remark that the Admin wants to process 3. The system displays the content of the remark and prompts the Admin to DELETE, REJECT, or CANCEL 4. If the Admin selects APPROVE, the system will set the remark's status to 'approved'.
Alternative Flows:	AF-S1: If Admin selects REJECT 1. If the Admin selects REJECT, the system will set the remark's status to 'rejected'. AF-S2: If Admin selects CANCEL 1. Admin can select CANCEL at any time to cancel the operation and the system will return to the SelectAdminActivity
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

I.V ModerateDiscussion

Use Case ID:	#1-5		
Use Case Name:	ModerateDiscussion		
Created By:	Tang Xinyue	Last Updated By:	Tang Xinyue
Date Created:	3rd September 2025	Date Last Updated:	15st November 2025

Actor:	Admin
Description:	Allows Admin to moderate pending discussion threads and decide whether to approve or reject them.
Preconditions:	Admin is logged in and authenticated.
Postconditions:	The discussion's status is set to "approved" or "rejected.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The ResaleSense lists the remarks that have been sent. 2. Admin selects the remark that the Admin wants to process 3. The system displays the content of the remark and prompts the Admin to DELETE, REJECT, or CANCEL 4. If the Admin selects APPROVE, the system will set the remark's status to 'approved'.
Alternative Flows:	AF-S1: If Admin selects REJECT 2. If the Admin selects REJECT, the system will set the remark's status to 'rejected'. AF-S2: If Admin selects CANCEL 2. Admin can select CANCEL at any time to cancel the operation and the system will return to the SelectAdminActivity
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

I.VI Flag User

Use Case ID:	#1-6		
Use Case Name:	FlagUser		
Created By:	Tang Xinyue	Last Updated By:	Tang Xinyue

Date Created:	3rd September 2025	Date Last Updated:	15st November 2025

Actor:	Admin
Description:	Allows Admin to flag a User for misconduct. This system follows a two-strike policy: the first flag is a warning; the second flag results in an automatic 7-day ban.
Preconditions:	Admin is logged in and authenticated
Postconditions:	User account status is updated to "flagged" (1st strike) or "banned" (2nd strike). A warning is added to the user's warnings array.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. Admin selects the "Flag Consumer" activity. 2. The system prompts Admin to enter the user ID and a mandatory reason. 3. Admin confirms the action. 4. The system checks the user's current status. 5. Response (First Strike): If status is "active," the system updates status to "flagged" and saves the reason to the user's warnings array. 6. Response (Second Strike): If status is already "flagged," the system updates status to "banned," sets banDuration to "7-days," and saves the reason. 7. The system provides a confirmation message.
Alternative Flows:	AF-1: If Admin selects *CANCEL*, the system returns to the SelectAdminActivity.
Exceptions:	EX-1: If the user ID does not exist, the system displays an error message.
Includes:	None
Special Requirements:	Reason for flagging must be mandatory.
Assumptions:	Flagging does not automatically ban; it triggers further monitoring.
Notes and Issues:	None

II. For Functional Requirement #2

II.I CustomerServiceSupport

Use Case ID:	#2-1		
Use Case Name:	CustomerServiceSupport		
Created By:	Shuma	Last Updated By:	Shuma
Date Created:	3rd September 2025	Date Last Updated:	15st November 2025

Actor:	User
Description:	Allows Users to communicate through in-app messaging for Customer Service Support services
Preconditions:	User must be logged in and authenticated
Postconditions:	None
Priority:	High
Frequency of Use:	High
Flow of Events:	1. User navigates to the Customer Service Support screen. 2. System shows a FAQ/help center, when scrolling to the bottom and clicking start live chat. 3. System displays the chat interface and a greeting message. 4. User selects to send a text message to Admin.
Alternative Flows:	None
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

III. For Functional Requirement #3

III.I QueryResale

Use Case ID:	#3-1		
Use Case Name:	Query Resale		
Created By:	Li Hang	Last Updated By:	Li Hang
Date Created:	1st September 2025	Date Last Updated:	15st November 2025

Actor:	User
Description:	Allows a User to find properties by applying filters via a search interface, or alternatively by browsing an interactive map, and then viewing the detailed insights for a selection.
Preconditions:	None
Postconditions:	The User is navigated to the Detailed Insights View for their selected property or area.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The system allows the User to view properties or areas by either through an interactive map or a search and filter interface. 2. The User selects to use the search and filter interface. 3. The User enters their desired criteria, such as location, room type, remaining lease duration (using a slider), price range, and house type. 4. The User applies the filters to execute the search. 5. The system displays a list of properties and/or areas that match the criteria. 6. The User selects a specific property or a designated area from the search results. 7. The system uses the included use case View Detailed Insights to display the analysis page for the selection.
Alternative Flows:	AF-S2: User chooses to use the interactive map instead 1. Instead of using the search interface, the User chooses to interact directly with the map. 2. The User visually navigates the map by panning and zooming to an area of interest. 3. The User selects a property pin or a designated area icon directly from the map. 4. The flow proceeds to Step 6 of the main flow.
Exceptions:	If the User's filter query results in no matching properties, the system shall display a user-friendly message indicating that no results were found.
Includes:	1. View Detailed Insights

Special Requirements:	None
Assumptions:	The data required for all the insights (historical, BTO, amenities) exists in the database.
Notes and Issues:	None

III.II ViewDetailedInsights

Use Case ID:	#3-2		
Use Case Name:	View Detailed Insights		
Created By:	Li Hang	Last Updated By:	Li Hang
Date Created:	1st September 2025	Date Last Updated:	15st November 2025

Actor:	User
Description:	Displays a detailed analysis page tailored to a user's selection, showing specific property analysis or aggregated area data.
Preconditions:	The User has successfully selected a property or an area from the map or a search result.
Postconditions:	The User is presented with the Detailed Insights View containing the relevant information for their selection.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. This use case begins after a User selects a specific property from the map or a search. 2. The system retrieves the data for the selected property. 3. The system displays the property-specific analysis page, containing its price history graph, future value prediction, valuation status, nearby amenities, Share/Bookmark buttons, and a "Data last updated on" timestamp.
Alternative Flows:	AF-S1: User selects a designated area instead of a specific property 1. The system retrieves the aggregated data for the selected area. 2. The system displays the area-specific analysis page, containing the area's average price trend, a list of available properties, the community discussion forum, and a "Data last updated on" timestamp.
Exceptions:	If the system fails to retrieve the necessary data for the selected property or area, it shall display a user-friendly error message.
Includes:	None
Special Requirements:	None
Assumptions:	The data required for all the insights (historical, amenities) exists in the database.

Notes and Issues:	This use case is typically included by a search or query use case and does not stand alone.

III.III BookmarkProperty

Use Case ID:	#3-3		
Use Case Name:	Bookmark Property		
Created By:	Li Hang	Last Updated By:	Li Hang
Date Created:	1st September 2025	Date Last Updated:	15st November 2025

Actor:	User
Description:	An optional extension that allows a User to save a specific property to their personal "My Bookmarks" list.
Preconditions:	The User is viewing a specific property in the View Detailed Insights use case. User must be verified.
Postconditions:	1. The property is added to or removed from the User's bookmarks list. 2. The bookmark icon on the page updates its state (e.g., filled or unfilled).
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. From the View Detailed Insights page, the User selects the "Bookmark" button for an unsaved property. 2. The system saves the property to the User's profile. 3. The system provides visual feedback (e.g., the icon fills in).
Alternative Flows:	AF-S1: User removes a bookmark 1. If the User selects the "Bookmark" button on an already-saved property, the system removes the property from the User's bookmark list. 2. The icon reverts to its "unsaved" state.
Exceptions:	If the save/remove action fails, an error message is displayed. If the user is not verified, the system shall display an error message
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	This use case extends View Detailed Insights.

III.IV SharePropertyInsights

Use Case ID:	#3-4		
Use Case Name:	Share Property Insights		
Created By:	Li Hang	Last Updated By:	Li Hang
Date Created:	1st September 2025	Date Last Updated:	15st November 2025

Actor:	User
Description:	An optional extension that allows a User to generate a unique, shareable link to the Detailed Insights page of a specific property.
Preconditions:	The User is viewing a specific property in the View Detailed Insights use case.
Postconditions:	A unique URL for the property page is made available to the User for sharing.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. From the View Detailed Insights page, the User selects the "Share" button. 2. The system generates a unique URL for the current page. 3. The system copies the link to the clipboard. 4. The system displays a confirmation message (e.g., "Link copied to clipboard")
Alternative Flows:	None
Exceptions:	If the native sharing functionality fails to launch, an appropriate message is displayed.
Includes:	None
Special Requirements:	May require integration with the device's native operating system for the share sheet.
Assumptions:	None
Notes and Issues:	This use case extends View Detailed Insights.

III.V StartDiscussion

Use Case ID:	#3-5		
Use Case Name:	Start Discussion		
Created By:	Izzat	Last Updated By:	Li Hang

Date Created:	31st August 2025	Date Last Updated:	15st November 2025

Actor:	User
Description:	An optional extension that allows a User to participate in the community discussion for a designated area by starting a new thread or commenting on an existing one.
Preconditions:	1. The User must be logged in and verified. 2. The User is viewing a specific property or area in the View Detailed Insights use case
Postconditions:	The system displays the User's comment
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. If a Thread does not exist on a listing, the system displays a "Start a Thread" button for users to create a thread on a particular listing. 2. When a User selects the "Start a Thread" button, the system will prompt the User to enter their text/comment. 3. The system will then prompt the user to confirm their text/comment via a "Submit Comment" Button. 4. If the User selects the "Submit Comment" Button, the system will save the text/comment in the database and display that it is submitted. 5. The system will then display the text/comment on that listing.
Alternative Flows:	If there has already been a Thread created on a particular listing: 1. The system will not display a "Start a Thread" button. 2. The system will only display the "Comment" button for Users to submit a comment on the listing.
Exceptions:	If any required information is missing or invalid, an error message is displayed. If the user is not verified, the system shall display an error message
Includes:	None
Special Requirements:	System needs to validate user input data.
Assumptions:	None
Notes and Issues:	None

III.VI EditDiscussion

Use Case ID:	#3-6
Use Case Name:	Edit Discussion

Created By:	Izzat	Last Updated By:	Li Hang
Date Created:	31st August 2025	Date Last Updated:	15st November 2025

Actor:	User
Description:	An optional extension that allows a User to edit or remove their previously submitted remarks on a property.
Preconditions:	1. The User must be logged in and verified. 2. The User is viewing a specific property or area in the View Detailed Insights use case. 3.The User has already submitted a remark on this property.
Postconditions:	The Remark is successfully edited.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. If a User has already made a remark on a particular listing, the system will display an “Edit” button beside their remark. 2. When the User selects the “Edit” button, the system will prompt the User to remove remarks or modify the remark text. 3. The system will display a “Save” button to allow the User to save the changes made, and a “Cancel” button to allow the User to abandon the changes made. 4. When the “Save” button is selected, the system will save the changes made
Alternative Flows:	None
Exceptions:	If any required information is missing or invalid, an error message is displayed.
Includes:	None
Special Requirements:	System needs to validate user input data.
Assumptions:	None
Notes and Issues:	This use case extends View Detailed Insights.

III.VII Submit Community Valuation

Use Case ID:	#3-7

Use Case Name:	Submit Community Valuation		
Created By:	Li Hang	Last Updated By:	Li Hang
Date Created:	11th September 2025	Date Last Updated:	15st November 2025

Actor:	User
Description:	An optional extension that allows a User to vote on the system's predicted fair value for a property and optionally provide a text remark to justify their vote.
Preconditions:	1. The User must be logged in and verified. 2. The User is viewing a specific property or area in the View Detailed Insights use case.
Postconditions:	1. The User's vote is anonymously added to the community consensus data. 2. If a remark is submitted, it is sent to the Admin moderation queue.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. From the ViewDetailedInsights page, the verified User clicks one of the vote options ("Over-valued," "Fair Value," or "Under-valued"). a. If the user is not verified, the system shows an alert. b. If the user has already voted, the buttons are disabled. 2. The system navigates the Consumer to the CreateRemark page. 3.The system displays a slider for the User to choose their suggested fair value and a free-text input field . 4.The User enters their text, adjusts the slider, and selects "Submit Remark." 5. The system sends the remark (containing text and communityValuation) to the Admin moderation queue and displays a confirmation.
Alternative Flows:	None
Exceptions:	1. If the vote fails to record due to a system error, an error message is displayed.
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	This use case extends View Detailed Insights.

IV. For Functional Requirement #4

IV.I CreateAccount

Use Case ID:	#4-1		
Use Case Name:	CreateAccount		
Created By:	Izzat	Last Updated By:	Izzat
Date Created:	31st August 2025	Date Last Updated:	15th November 2025

Actor:	User
Description:	Allows a person to create an Account in ResaleSense to become a Consumer.
Preconditions:	None
Postconditions:	An Account with the role of Consumer is created for the User.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The system prompts the User to enter their username, password, email. 2. The User enters the required information and selects the “Sign Up" button to confirm the inputs. 3. An Account with the 'Consumer' role is created for the User.
Alternative Flows:	If the User selects the back navigation icon, the system will navigate the User to the login page.
Exceptions:	1. If any required information is missing or invalid (password is not at least 6 characters long), an error message is displayed. 2. If the name or email is already in use, a 409 error is returned.
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

IV.II Login

Use Case ID:	#4-2		
Use Case Name:	Login		
Created By:	Izzat	Last Updated By:	Izzat
Date Created:	31st August 2025	Date Last Updated:	15th November 2025

Actor:	User
Description:	Allows the User to log into their ResaleSense Account using their email and password.
Preconditions:	None
Postconditions:	User is logged into the ResaleSense website and is navigated to the home screen of the website.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The system allows the User to login with Email and password 2. The User chooses to login with email and password. 3. The User enters his/her email and password. The password is masked as dots, but the User can choose to unmask it by clicking on the eye icon. 4. The User selects the “Login” button.
Alternative Flows:	None
Exceptions:	1. If any required information is missing or invalid, an error message is displayed. 2. If email and password do not match when the User tries to login, ResaleSense shall display “Email and password do not match” to the user.
Includes:	None
Special Requirements:	System needs to validate user input data
Assumptions:	The User has an existing ResaleSense Account.
Notes and Issues:	None